Week 04 teacher guide

Dividing fractions

Keep this guide beside PowerPoint slides 110-147. Follow the numbered slide sequence, then use the matching answer section after the student has tried.

The seven days

Day 022. Saturday | pages 89-95 | slides 110-127
The meaning of fraction division

Day 023. Sunday | pages 96-105 | slides 128-147
Fraction division and reciprocals

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Dividing fractions

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Dividing fractions

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Weekly check

This week's idea

Division asks how many divisor-sized groups fit. Equal small units make the comparison visible. The reciprocal rule records that reasoning efficiently.

Use the built-in Booster review where it appears in the day's work. It replaces ordinary practice. No separate Booster packet is needed.

PDF bookmarks jump to each lesson and answer section. The student workbook's Sunday reference pages are for checking an example after an attempt. Keep this teacher guide and the answer-review slides out of view during independent work.

Saturday. The meaning of fraction division

Day 022, pages 89-95 | PowerPoint slides 110-127

Goal: I can show how many fractional groups fit in an amount.

Set out: fraction strips, blank number lines, pencil.

Check first: Fraction equivalence and division by unit fractions.

Pacing: Retrieval 5; launch 5; explore 15; discuss 10; synthesis 5; exit 5; preview 5; slack 10. Total 60 min.

Explore: what to watch for

Labels what counts as one group. Why it matters: Fractional quotients refer to the divisor-sized group. Tutor move: Ask the student to circle one whole group.

Renames both amounts using the same small unit. Why it matters: Counting comparable pieces makes division visible. Tutor move: Accept a tape or a number line with equal units.

Calls the remainder a fraction of the group. Why it matters: Avoids treating leftover units as the answer fraction. Tutor move: Compare the leftover pieces with pieces in one group.

Likely misconceptions

Reverses dividend and divisor. Why: Has not named the available amount. Tutor move: Label amount first and group size second.

Multiplies the fractions. Why: Applies a rule before making sense of division. Tutor move: Count equal pieces before writing a rule.

Follow the slide numbers below. Let the student attempt the task before showing the model or the answer. Write on the matching workbook pages; use extra paper only when more room is helpful.

Slide 110. Week 4, Saturday

Topic: The meaning of fraction division
Student: Day 022, pages 89-95

Say: I can show how many fractional groups fit in an amount.

Ask: What do you already know about this idea?

Expect: Fraction equivalence and division by unit fractions.

If stuck: Ask the student to circle one whole group.

Slide 111. Today

Topic: The meaning of fraction division
Student: Day 022, pages 89-95

Say: We will leave time to explain and check.

Ask: What can a model help us see?

Expect: The divisor names one group. A quotient tells how many of those groups fit, even when it is part of a group.

If stuck: Read the learning goal together.

Slide 112. Recall | 3 min

Topic: The meaning of fraction division
Student: Day 022, pages 89-95

Say: Work from memory. We will check together.

Ask: Which earlier model helps?

Expect: 1: $\frac{7}{8}$; 2: 20 blue tiles; 3: 0.65 dollars per marker; 4: 20; $\frac{1}{20}$

If stuck: Give time to draw before giving a hint.

Slide 113. Recall: answers | 2 min

Topic: The meaning of fraction division
Student: Day 022, pages 89-95

Say: Check one answer at a time.

Ask: Which step would you change?

Expect: $\frac{7}{8}$; 20 blue tiles; 0.65 dollars per marker; 20; $\frac{1}{20}$

If stuck: Rebuild one missed item with its model.

Slide 114. Launch | 5 min

Topic: The meaning of fraction division
Student: Day 022, pages 89-95

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Agree or disagree: A quotient below one means the division cannot be done.

Expect: P1: 20 pieces

If stuck: Ask the student to circle one whole group.

Slide 115. Explore | 3 min

Topic: The meaning of fraction division
Student: Day 022, pages 89-95

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Agree or disagree: A quotient below one means the division cannot be done.

Expect: P2: 6 scoops; P3: 6 pieces

If stuck: Ask the student to circle one whole group.

Slide 116. The model | 3 min

Topic: The meaning of fraction division
Student: Day 022, pages 89-95

Say: Match each part of the model to a number in the problem.

Ask: What does one part represent?

Expect: Five sixths is five fourths of a four-sixths group.

If stuck: Ask the student to circle one whole group.

Slide 117. Required practice | 3 min

Topic: The meaning of fraction division
Student: Day 022, pages 89-95

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Agree or disagree: A quotient below one means the division cannot be done.

Expect: P4: 1 $\frac{1}{4}$ groups; P5: $\frac{2}{5}$ of a scoop

If stuck: Ask the student to circle one whole group.

Slide 118. Required practice | 2 min

Topic: The meaning of fraction division
Student: Day 022, pages 89-95

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Agree or disagree: A quotient below one means the division cannot be done.

Expect: P6: $\frac{9}{10}$ liter; P7: 1 $\frac{1}{2}$ groups. Stories vary.

If stuck: Ask the student to circle one whole group.

Slide 119. Watch the thinking | 2 min

Topic: The meaning of fraction division
Student: Day 022, pages 89-95

Say: Read the reason beside each step.

Ask: Where does the model appear in the calculation?

Expect: 1 $\frac{1}{2}$

If stuck: Return to the matching unit in the model.

Slide 120. Your turn | 2 min

Topic: The meaning of fraction division
Student: Day 022, pages 89-95

Say: Complete the unfinished method.

Ask: Which step is still needed?

Expect: 2 eighths; 2 $\frac{1}{2}$ groups

If stuck: Compare with the previous example without copying its numbers.

Slide 121. Compare methods | 5 min

Topic: The meaning of fraction division
Student: Day 022, pages 89-95

Say: Partner compare: show the same problem in two ways.

Ask: Agree or disagree: A quotient below one means the division cannot be done. Partner compare: Why is the leftover one fourth in the example half of a group?

Expect: Accept drawings and equations that preserve the same quantities and units. Renames both amounts using the same small unit.

If stuck: Accept a tape or a number line with equal units.

Slide 122. A claim to check | 5 min

Topic: The meaning of fraction division
Student: Day 022, pages 89-95

Say: Agree or disagree. Find the first step that fails.

Ask: Why does this mistake seem reasonable? How can a model check it?

Expect: It asks what fraction of a five-sixths group fits in one third. The answer is $\frac{2}{5}$.

If stuck: Rename one third as two sixths and compare with five sixths.

Slide 123. Words and meaning | 3 min

Topic: The meaning of fraction division
Student: Day 022, pages 89-95

Say: Use each word while pointing to a model.

Ask: Which example shows the meaning most clearly?

Expect: In $2 \div \frac{3}{4}$, the dividend is 2. In $2 \div \frac{3}{4}$, one group is $\frac{3}{4}$. $\frac{3}{4} \div \frac{1}{4} = 3$ groups.

If stuck: Read one definition and draw its example.

Slide 124. In your own words | 2 min

Topic: The meaning of fraction division
Student: Day 022, pages 89-95

Say: Use your own words and one example.

Ask: Agree or disagree: A quotient below one means the division cannot be done.

Expect: The divisor names one group. A quotient tells how many of those groups fit, even when it is part of a group.

If stuck: Point to the model and name what each part represents.

Slide 125. Exit ticket | 5 min

Topic: The meaning of fraction division
Student: Day 022, pages 89-95

Say: Work independently. Include the units and the requested reasoning.

Ask: Read the problem once. What is it asking you to find?

Expect: $\frac{8}{15}$ of the length

If stuck: Do not coach during the ticket. Use the sort after collecting it.

Slide 126. The week ahead | 5 min

Topic: The meaning of fraction division
Student: Day 022, pages 89-95

Say: Each packet is a short practice session.

Ask: Which model might help you during the week?

Expect: A model connected to the current topics.

If stuck: Open the study guide to the matching example.

Slide 127. Tutor reference

Topic: The meaning of fraction division
Student: Day 022, pages 89-95

Tutor only. Keep this slide hidden during the lesson. Use it after collecting the exit ticket.

Exit answer: $\frac{8}{15}$ of the length

Got it: Gives $\frac{8}{15}$ and shows 8 equal units compared with a 15-unit group.

Almost there: Correct quotient but an unclear drawing, or a correct comparison model with a simplification slip.

Needs re-teaching: Multiplies, reverses the fractions, or says the answer must exceed one.

Sunday. Fraction division and reciprocals

Day 023, pages 96-105 | PowerPoint slides 128-147

Goal: I can connect the reciprocal rule to equal groups and use it in measurement problems.

Set out: fraction strips, grid paper, small box.

Check first: Fraction division with equal-part models and mixed-number conversion.

Pacing: Retrieval 5; launch 5; explore 12; discuss 10; synthesis 3; mixed review 10; exit 5; preview 5; slack 5. Total 60 min.

Explore: what to watch for

Converts the mixed number before dividing. Why it matters: Preserves the total amount. Tutor move: Ask how many eighths make both whole yards.

Uses a model or reciprocal and checks by multiplication. Why it matters: Both methods are valid and connected. Tutor move: Accept either method and ask what the quotient counts.

Uses square units for area and cubic units for volume. Why it matters: The dimensions measure different things. Tutor move: Trace one face, then point inside the box.

Likely misconceptions

Inverts the dividend. Why: Memorizes a rule without a division meaning. Tutor move: Check quotient × divisor = dividend.

Leaves out the whole-number part. Why: Treats a mixed number as only its fraction. Tutor move: Build the whole amount with equal fractional pieces.

Follow the slide numbers below. Let the student attempt the task before showing the model or the answer. Write on the matching workbook pages; use extra paper only when more room is helpful.

Slide 128. Week 4, Sunday

Topic: Fraction division and reciprocals
Student: Day 023, pages 96-105

Say: I can connect the reciprocal rule to equal groups and use it in measurement problems.

Ask: What do you already know about this idea?

Expect: Fraction division with equal-part models and mixed-number conversion.

If stuck: Ask how many eighths make both whole yards.

Slide 129. Today

Topic: Fraction division and reciprocals
Student: Day 023, pages 96-105

Say: We will leave time to explain and check.

Ask: What can a model help us see?

Expect: Dividing by a fraction counts groups. Multiplying by its reciprocal gives the same count.

If stuck: Read the learning goal together.

Slide 130. Recall | 3 min

Topic: Fraction division and reciprocals
Student: Day 023, pages 96-105

Say: Work from memory. We will check together.

Ask: Which earlier model helps?

Expect: 1: $\frac{5}{6}$; 2: 12; 3: 75 %; 4: 12 pieces

If stuck: Give time to draw before giving a hint.

Slide 131. Recall: answers | 2 min

Topic: Fraction division and reciprocals
Student: Day 023, pages 96-105

Say: Check one answer at a time.

Ask: Which step would you change?

Expect: $\frac{5}{6}$; 12; 75 %; 12 pieces

If stuck: Rebuild one missed item with its model.

Slide 132. Launch | 5 min

Topic: Fraction division and reciprocals
Student: Day 023, pages 96-105

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Agree or disagree: In a division problem, either fraction can be inverted. Test the claim with a tape.

Expect: P1: $\frac{3}{20}$ liter

If stuck: Ask how many eighths make both whole yards.

Slide 133. Explore | 2 min

Topic: Fraction division and reciprocals
Student: Day 023, pages 96-105

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Agree or disagree: In a division problem, either fraction can be inverted. Test the claim with a tape.

Expect: P2: 6 $\frac{2}{3}$ groups; P3: 3 $\frac{1}{2}$

If stuck: Ask how many eighths make both whole yards.

Slide 134. The model | 2 min

Topic: Fraction division and reciprocals
Student: Day 023, pages 96-105

Say: Match each part of the model to a number in the problem.

Ask: What does one part represent?

Expect: One whole has $\frac{5}{3}$ groups of size $\frac{3}{5}$.

If stuck: Ask how many eighths make both whole yards.

Slide 135. Required practice | 2 min

Topic: Fraction division and reciprocals
Student: Day 023, pages 96-105

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Agree or disagree: In a division problem, either fraction can be inverted. Test the claim with a tape.

Expect: P4: 3; P5: 4 pieces

If stuck: Ask how many eighths make both whole yards.

Slide 136. Required practice | 2 min

Topic: Fraction division and reciprocals
Student: Day 023, pages 96-105

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Agree or disagree: In a division problem, either fraction can be inverted. Test the claim with a tape.

Expect: P6: $1\ \frac{1}{4}$ meters

If stuck: Ask how many eighths make both whole yards.

Slide 137. Required practice | 2 min

Topic: Fraction division and reciprocals
Student: Day 023, pages 96-105

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Agree or disagree: In a division problem, either fraction can be inverted. Test the claim with a tape.

Expect: P7: 1 cubic meter

If stuck: Ask how many eighths make both whole yards.

Slide 138. Watch the thinking | 1 min

Topic: Fraction division and reciprocals
Student: Day 023, pages 96-105

Say: Read the reason beside each step.

Ask: Where does the model appear in the calculation?

Expect: $1\ \frac{1}{9}$

If stuck: Return to the matching unit in the model.

Slide 139. Your turn | 1 min

Topic: Fraction division and reciprocals
Student: Day 023, pages 96-105

Say: Complete the unfinished method.

Ask: Which step is still needed?

Expect: $\frac{3}{2}$; $1\ \frac{1}{5}$

If stuck: Compare with the previous example without copying its numbers.

Slide 140. Compare methods | 5 min

Topic: Fraction division and reciprocals
Student: Day 023, pages 96-105

Say: Partner compare: show the same problem in two ways.

Ask: Agree or disagree: In a division problem, either fraction can be inverted. Test the claim with a tape. Partner compare: How does finding a missing rectangle side use the same division idea as cutting ribbon?

Expect: Accept drawings and equations that preserve the same quantities and units. Uses a model or reciprocal and checks by multiplication.

If stuck: Accept either method and ask what the quotient counts.

Slide 141. A claim to check | 5 min

Topic: Fraction division and reciprocals
Student: Day 023, pages 96-105

Say: Agree or disagree. Find the first step that fails.

Ask: Why does this mistake seem reasonable? How can a model check it?

Expect: Keep the dividend $\frac{7}{8}$. Multiply by the reciprocal of the divisor: $\frac{7}{8} \times 4 = 3\ \frac{1}{2}$.

If stuck: Check whether $\frac{2}{7}$ groups of one fourth could make seven eighths.

Slide 142. Words and meaning | 2 min

Topic: Fraction division and reciprocals
Student: Day 023, pages 96-105

Say: Use each word while pointing to a model.

Ask: Which example shows the meaning most clearly?

Expect: $\frac{3}{5}$ and $\frac{5}{3}$ are reciprocals. $2\ \frac{1}{3} = \frac{7}{3}$. A box uses cubic units.

If stuck: Read one definition and draw its example.

Slide 143. Mixed review | 10 min

Topic: Fraction division and reciprocals
Student: Day 023, pages 96-105

Say: Choose a method for each problem.

Ask: What tells you which operation to use?

Expect: 2; 3 cups; 0.65 dollars per marker

If stuck: Name the quantities and units before calculating.

Slide 144. In your own words | 1 min

Topic: Fraction division and reciprocals
Student: Day 023, pages 96-105

Say: Use your own words and one example.

Ask: Agree or disagree: In a division problem, either fraction can be inverted. Test the claim with a tape.

Expect: Dividing by a fraction counts groups. Multiplying by its reciprocal gives the same count.

If stuck: Point to the model and name what each part represents.

Slide 145. Exit ticket | 5 min

Topic: Fraction division and reciprocals
Student: Day 023, pages 96-105

Say: Work independently. Include the units and the requested reasoning.

Ask: Read the problem once. What is it asking you to find?

Expect: 7 pieces

If stuck: Do not coach during the ticket. Use the sort after collecting it.

Slide 146. The week ahead | 5 min

Topic: Fraction division and reciprocals
Student: Day 023, pages 96-105

Say: Each packet is a short practice session.

Ask: Which model might help you during the week?

Expect: A model connected to the current topics.

If stuck: Open the study guide to the matching example.

Slide 147. Tutor reference

Topic: Fraction division and reciprocals

Student: Day 023, pages 96-105

Tutor only. Keep this slide hidden during the lesson. Use it after collecting the exit ticket.

Exit answer: 7 pieces

Got it: Finds 7 pieces, converts to $\frac{21}{8}$, and shows seven groups or a valid reciprocal calculation.

Almost there: Uses the correct mixed-number conversion and division setup but makes a small arithmetic slip.

Needs re-teaching: Drops the 2 whole yards, multiplies by $\frac{3}{8}$, or inverts the dividend.

Weekday teaching and review

Use one day at a time. Aim for 15-20 minutes, or 15 minutes during a holiday week. Accuracy and a clear explanation matter more than finishing quickly.

Day 024. Monday | pages 106-107

Ask for a model from Saturday before checking an answer. Let the student choose a method. After checking, have the student correct the first step that changed the quantity.

Built-in Booster review: Fraction operations: items 9, 10, 11, 12. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 025. Tuesday | pages 108-109

Connect the practice to Sunday's model. Ask what each number counts and how a multiplication check, diagram, or estimate supports the answer.

Built-in Booster review: Fraction operations: items 9, 10, 11, 12. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 026. Wednesday | pages 110-111

Let the student choose the method for each mixed problem. For the puzzle, ask which conditions are already met. Accept any solution that meets every condition.

Built-in Booster review: Weekly puzzle: items 13. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 027. Thursday | pages 112-113

Ask the student to explain the planted error before repairing it. Optional Stretch comes after the required work and may be skipped without delaying the next day.

Built-in Booster review: Percent practice: items 9, 10, 11, 12. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 028. Friday | pages 114-115

Give the weekly check independently, without coaching or a calculator. Score the required reasoning as well as the answer. Afterward, choose one error to discuss calmly.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

After Friday

On a 10-item check: 9-10 correct is strong; 7-8 is on track; 5-6 needs another review of the missed topic; 0-4 calls for 20 minutes of re-teaching at the next session. Record the skill and one observation. Check it again with a fresh item at least a week later.

Complete answers for the week

Match the Day number, item ID, and student page before checking. Questions stay in the student workbook; these keys give every answer and the needed reasoning. Equivalent correct forms and sound alternative methods are welcome. An explanation item needs a reason, model, or valid check as well as its conclusion.

The meaning of fraction division

Day 022, student workbook pages 89-93

R1 [F-COMP] | student p. 89

Answer: $\frac{7}{8}$

Reasoning: The gaps from 1 are one eighth and one fifth. One eighth is smaller.

R2 [R-EQUIV] | student p. 89

Answer: 20 blue tiles

Reasoning: The number of batches is 12 ÷ 3. Multiply that count by 5.

Watch for 14 blue tiles: Adding the same amount to both counts changes the ratio.

R3 [R-RATE] | student p. 89

Answer: 0.65 dollars per marker

Reasoning: Use the named units, calculate, then check with the inverse operation.

R4 [F-DIV] | student p. 89

Answer: 20; $\frac{1}{20}$

Reasoning: The first counts fifths in four. The second splits one fifth into four shares.

Watch for $\frac{4}{5}$; $\frac{4}{5}$: Changing the order changes what is shared or counted.

S [EXPLAIN] | student p. 89

Answer: Answers vary. The divisor names one group. A quotient tells how many of those groups fit, even when it is part of a group.

Reasoning: Accept an accurate explanation supported by a matching example.

P1 [F-DIV] | student p. 90

Answer: 20 pieces

Reasoning: Each meter contains five fifths. Four meters contain twenty.

P2 [F-DIV] | student p. 90

Answer: 6 scoops

Reasoning: Four cups are twelve thirds. Each scoop uses two thirds, so twelve thirds make six scoops.

P3 [F-DIV] | student p. 90

Answer: 6 pieces

Reasoning: Three fourths is six eighths. Count six pieces.

P4 [F-DIV] | student p. 91

Answer: $1\frac{1}{4}$ groups

Reasoning: One group is four sixths. Five sixths is five fourths of that group.

P5 [F-DIV] | student p. 91

Answer: $\frac{2}{5}$ of a scoop

Reasoning: One scoop has five sixths. One third is two sixths, so it fills two of five equal scoop parts.

P6 [F-DIV] | student p. 91

Answer: $\frac{9}{10}$ liter

Reasoning: Two thirds of the jar hold three fifths liter. One third holds three tenths, so three thirds hold nine tenths.

P7 [F-DIV] | student p. 92

Answer: $1\frac{1}{2}$ groups. Stories vary.

Reasoning: Example: A pitcher has three fourths liter. How many half-liter servings does it hold, including part of a serving?

F [F-DIV] | student p. 92

Answer: 2 eighths; $2\frac{1}{2}$ groups

Reasoning: Use the worked example in the slides to check each step.

D [F-DIV] | student p. 92

Answer: It asks what fraction of a five-sixths group fits in one third. The answer is $\frac{2}{5}$.

Reasoning: He uses a familiar fraction rule without identifying the group. Rename one third as two sixths and compare with five sixths.

E [F-DIV] | student p. 93

Answer: $\frac{8}{15}$ of the length

Reasoning: Use twentieths. The target length is 15 twentieths; the available amount is 8 twentieths. That is eight fifteenths of one group.

Watch for $\frac{3}{10}$: Multiplying the two fractions finds a different amount.

Fraction division and reciprocals

Day 023, student workbook pages 96-101

R1 [F-ADD] | student p. 96

Answer: $\frac{5}{6}$

Reasoning: Rename both fractions with a common denominator. Add the counts of equal parts.

R2 [R-TABLE] | student p. 96

Answer: 12

Reasoning: Multiply both entries in the first pair by the same factor.

R3 [P-PERCENT] | student p. 96

Answer: 75 %

Reasoning: Compare the green count with the total. Rename the fraction as a rate per hundred.

R4 [F-DIV] | student p. 96

Answer: 12 pieces

Reasoning: Each meter contains 4 equal pieces. Multiply by 3 meters.

Watch for $\frac{3}{4}$: Multiplying by the divisor answers a different question.

S [EXPLAIN] | student p. 96

Answer: Answers vary. Dividing by a fraction counts groups. Multiplying by its reciprocal gives the same count.

Reasoning: Accept an accurate explanation supported by a matching example.

P1 [F-DIV] | student p. 97

Answer: $\frac{3}{20}$ liter

Reasoning: Each fourth is split into five parts, so each bottle gets three twentieths.

P2 [F-DIV] | student p. 97

Answer: $6\ \frac{2}{3}$ groups

Reasoning: Four cups are twenty fifths. Each group uses three fifths. Count twenty thirds of a group.

P3 [F-DIV] | student p. 97

Answer: $3\ \frac{1}{2}$

Reasoning: One quarter is two eighths. Seven eighths contains seven halves of a two-eighths group.

P4 [F-DIV] | student p. 98

Answer: 3

Reasoning: Rename $1\ \frac{7}{8}$ as $\frac{15}{8}$. Three groups of five eighths fit.

P5 [F-DIV] | student p. 98

Answer: 4 pieces

Reasoning: Convert $3\ \frac{1}{2}$ to $\frac{28}{8}$. Four groups of seven eighths fit.

P6 [F-DIV] | student p. 98

Answer: $1\ \frac{1}{4}$ meters

Reasoning: Length × width = area. Length is $\frac{5}{6} \div \frac{2}{3}$.

P7 [G-VOL] | student p. 99

Answer: 1 cubic meter

Reasoning: Multiply the three perpendicular edge lengths. The base area is $\frac{1}{2}$ square meter; two layers make one cubic meter.

F [F-DIV] | student p. 99

Answer: $\frac{3}{2}$; $1\ \frac{1}{5}$

Reasoning: Use the worked example in the slides to check each step.

D [F-DIV] | student p. 99

Answer: Keep the dividend $\frac{7}{8}$. Multiply by the reciprocal of the divisor: $\frac{7}{8} \times 4 = 3\ \frac{1}{2}$.

Reasoning: She reverses the dividend instead of changing the size of the divisor-sized group. Check whether $\frac{2}{7}$ groups of one fourth could make seven eighths.

M1 [F-MULT] | student p. 100

Answer: 2

Reasoning: Count two thirds for each whole-number group, then rename the result.

M2 [R-EQUIV] | student p. 100

Answer: 3 cups

Reasoning: Find the number of recipe batches by dividing the rice amount by 3. Each batch needs 2 cups beans.

M3 [R-RATE] | student p. 100

Answer: 0.65 dollars per marker

Reasoning: Use the named units, calculate, then check with the inverse operation.

E [F-DIV] | student p. 101

Answer: 7 pieces

Reasoning: $2\ \frac{5}{8} = \frac{21}{8}$. Twenty-one eighths make seven groups of three eighths.

Watch for $\frac{7}{8}$ or $\frac{1}{7}$: Dropping the whole part or reversing the wrong fraction changes the total.

Study guide: Try it

Day 023, student workbook pages 102-105

1 [F-DIV] | student p. 105

Answer: $1\ \frac{1}{6}$ groups

Reasoning: A group is six eighths. Seven eighths is seven sixths of that group.

2 [F-DIV] | student p. 105

Answer: $\frac{8}{9}$ liter

Reasoning: One fourth holds two ninths liter. Four fourths hold eight ninths liter.

3 [F-DIV] | student p. 105

Answer: $\frac{5}{24}$ liter

Reasoning: Split each sixth into four equal parts. Each cup gets five twenty-fourths.

4 [F-DIV] | student p. 105

Answer: 12 pieces

Reasoning: Rename the total as thirty-six eighths. Divide into groups of three eighths.

5 [F-DIV] | student p. 105

Answer: $1\ \frac{3}{7}$

Reasoning: Keep four sevenths and multiply by five halves. The check recovers four sevenths.

Daily practice

Day 024, student workbook pages 106-107

1 [N-FACT] | student p. 106

Answer: 18

2 [N-FACT] | student p. 106

Answer: 42

3 [N-FACT] | student p. 106

Answer: 24

4 [N-FACT] | student p. 106

Answer: 54

5 [F-DIV] | student p. 106

Answer: $\frac{2}{3}$

Reasoning: The available amount is smaller than one group. Multiply by the reciprocal of the group size.

Watch for $\frac{3}{8}$: Multiplying by the divisor answers a different question.

6 [F-DIV] | student p. 106

Answer: 2

Reasoning: Both fractions use the same unit. Count how many single parts fit.

Watch for $\frac{1}{8}$: Multiplying by the divisor answers a different question.

7 [F-DIV] | student p. 106

Answer: $\frac{3}{4}$

Reasoning: One half is three sixths. A group is four sixths. Three of four group parts fit.

Watch for $\frac{1}{3}$: The proposed answer multiplies instead of dividing.

8 [F-DIV] | student p. 106

Answer: $2\ \frac{1}{2}$; stories vary

Reasoning: Example: how many one-third-meter lengths fit in five sixths meter, allowing part of a length?

9 [F-DIV] | student p. 107 | Built-in review: Fraction operations

Answer: $\frac{5}{8}$

Reasoning: The available amount is smaller than one group. Multiply by the reciprocal of the group size.

Watch for $\frac{2}{5}$: Multiplying by the divisor answers a different question.

10 [F-DIV] | student p. 107 | Built-in review: Fraction operations

Answer: $\frac{10}{3}$

11 [F-DIV] | student p. 107 | Built-in review: Fraction operations

Answer: $\frac{5}{3}$

12 [F-DIV] | student p. 107 | Built-in review: Fraction operations

Answer: $\frac{25}{3}$

13 [F-DIV] | student p. 107

Answer: It asks what fraction of a five-sixths group fits in one third. The answer is $\frac{2}{5}$.

Watch for $\frac{5}{18}$: He uses a familiar fraction rule without identifying the group.

Daily practice

Day 025, student workbook pages 108-109

1 [F-EQUIV] | student p. 108

Answer: 0.5; 50%

2 [F-EQUIV] | student p. 108

Answer: 0.25; 25%

3 [F-EQUIV] | student p. 108

Answer: 0.75; 75%

4 [F-EQUIV] | student p. 108

Answer: 0.125; 12.5%

5 [F-DIV] | student p. 108

Answer: $4\ \frac{2}{3}$

Reasoning: Rename the mixed number as an improper fraction, then count groups of three fourths.

Watch for $2\ \frac{5}{8}$: Multiplying by the divisor answers a different question.

6 [F-DIV] | student p. 108

Answer: 12 pieces

Reasoning: Rename the total as thirty-six eighths. Divide into groups of three eighths.

7 [F-DIV] | student p. 108

Answer: $\frac{5}{24}$ liter

Reasoning: Split each sixth into four equal parts. Each cup gets five twenty-fourths.

8 [P-PART] | student p. 108

Answer: 42 points

Reasoning: Take all 30 points, then add forty percent, which is 12 points.

9 [F-DIV] | student p. 109 | Built-in review: Fraction operations

Answer: $\frac{2}{5}$

Reasoning: Two and one half is five halves. Five halves times two fifths equals one.

10 [F-DIV] | student p. 109 | Built-in review: Fraction operations

Answer: 2

Reasoning: Rename the mixed number as an improper fraction, then count groups of three fourths.

Watch for $1\ \frac{1}{8}$: Multiplying by the divisor answers a different question.

11 [F-DIV] | student p. 109 | Built-in review: Fraction operations

Answer: $\frac{20}{9}$

12 [F-MULT] | student p. 109 | Built-in review: Fraction operations

Answer: $\frac{16}{15}$

13 [F-DIV] | student p. 109

Answer: Keep the dividend $\frac{7}{8}$. Multiply by the reciprocal of the divisor: $\frac{7}{8} \times 4 = 3\ \frac{1}{2}$.

Watch for $\frac{2}{7}$: She reverses the dividend instead of changing the size of the divisor-sized group.

Daily practice

Day 026, student workbook pages 110-111

1 [N-SCALE] | student p. 110

Answer: 62

2 [N-SCALE] | student p. 110

Answer: 6

3 [N-SCALE] | student p. 110

Answer: 13

4 [N-SCALE] | student p. 110

Answer: 18

5 [F-DIV] | student p. 110

Answer: $4\ \frac{1}{2}$

Reasoning: Rename each whole in thirds, then count groups of two thirds.

Watch for 2: Multiplying by the divisor answers a different question.

6 [F-DIV] | student p. 110

Answer: $\frac{5}{8}$

Reasoning: The available amount is smaller than one group. Multiply by the reciprocal of the group size.

Watch for $\frac{2}{5}$: Multiplying by the divisor answers a different question.

7 [R-TABLE] | student p. 110

Answer: 16

Reasoning: Multiply both entries in the first pair by the same factor.

8 [R-EQUIV] | student p. 110

Answer: 5 cups

Reasoning: Find the number of recipe batches by dividing the rice amount by 3. Each batch needs 2 cups beans.

9 [F-ADD] | student p. 111

Answer: $\frac{5}{6}$

Reasoning: Rename both fractions with a common denominator. Add the counts of equal parts.

10 [F-MULT] | student p. 111

Answer: $\frac{3}{5}$

Reasoning: An area model shows 12 of 20 equal parts, or three fifths.

11 [N-FACT] | student p. 111

Answer: 8

12 [G-VOL] | student p. 111

Answer: 24 cm^3

13 [PUZZLE] | student p. 111 | Built-in review: Weekly puzzle

Answer: 3 one-third-liter scoops and 2 half-liter scoops

Reasoning: Try five thirds first: $\frac{5}{3}$ liters. Replacing a third with a half adds $\frac{1}{6}$ liter. Two replacements fill the missing third.

Daily practice

Day 027, student workbook pages 112-113

1 [N-EST] | student p. 112

Answer: 600

2 [N-EST] | student p. 112

Answer: 7

3 [N-EST] | student p. 112

Answer: About 100

4 [N-EST] | student p. 112

Answer: About 20

5 [F-DIV] | student p. 112

Answer: $4\frac{1}{2}$

Reasoning: Rename each whole in thirds, then count groups of two thirds.

Watch for 2: Multiplying by the divisor answers a different question.

6 [F-DIV] | student p. 112

Answer: $\frac{5}{8}$

Reasoning: The available amount is smaller than one group. Multiply by the reciprocal of the group size.

Watch for $\frac{2}{5}$: Multiplying by the divisor answers a different question.

7 [F-DIV] | student p. 112

Answer: $\frac{3}{4}$

Reasoning: One half is three sixths. A group is four sixths. Three of four group parts fit.

Watch for $\frac{1}{3}$: The proposed answer multiplies instead of dividing.

8 [R-GRAPH] | student p. 112

Answer: Yes

Reasoning: Every vertical value is twice the horizontal value.

9 [P-PART] | student p. 113 | Built-in review: Percent practice

Answer: 1.8000 dollars

Reasoning: Ten percent is $1.50 and two percent is $0.30. Add them.

10 [P-WHOLE] | student p. 113 | Built-in review: Percent practice

Answer: 60 shells

Reasoning: Divide the given amount into three 10% groups, then take ten such groups.

Watch for 5.40 shells: Taking 30% of the known part does not recover the whole.

11 [P-PERCENT] | student p. 113 | Built-in review: Percent practice

Answer: 75 %

Reasoning: Compare the green count with the total. Rename the fraction as a rate per hundred.

12 [P-PART] | student p. 113 | Built-in review: Percent practice

Answer: 10 stickers

Reasoning: Twenty-five percent is one fourth. Divide the whole by 4.

13 [F-DIV] | student p. 113

Answer: 1 meters

Reasoning: Divide area by width. Check that length times width equals area.

Watch for $\frac{9}{16}$: Multiplying by the divisor answers a different question.

14 [F-DIV] | student p. 113

Answer: Keep the dividend $\frac{7}{8}$. Multiply by the reciprocal of the divisor: $\frac{7}{8} \times 4 = 3\ \frac{1}{2}$.

Watch for $\frac{2}{7}$: She reverses the dividend instead of changing the size of the divisor-sized group.

Weekly check

Day 028, student workbook pages 114-115

1 [F-DIV] | student p. 114

Answer: $4\ \frac{1}{2}$

Reasoning: Rename each whole in thirds, then count groups of two thirds.

Watch for 2: Multiplying by the divisor answers a different question.

2 [F-DIV] | student p. 114

Answer: $\frac{5}{8}$

Reasoning: The available amount is smaller than one group. Multiply by the reciprocal of the group size.

Watch for $\frac{2}{5}$: Multiplying by the divisor answers a different question.

3 [F-DIV] | student p. 114

Answer: $\frac{3}{4}$

Reasoning: One half is three sixths. A group is four sixths. Three of four group parts fit.

Watch for $\frac{1}{3}$: The proposed answer multiplies instead of dividing.

4 [F-DIV] | student p. 114

Answer: $1\ \frac{2}{3}$ meters

Reasoning: Divide area by width. Check that length times width equals area.

Watch for $\frac{15}{16}$: Multiplying by the divisor answers a different question.

5 [R-RATE] | student p. 114

Answer: 0.65 dollars per marker

Reasoning: Use the named units, calculate, then check with the inverse operation.

6 [P-PART] | student p. 115

Answer: 4, 20, and 30 counters

Reasoning: Use one tenth, one half, and three fourths of the whole.

7 [R-TABLE] | student p. 115

Answer: 20

Reasoning: Multiply both entries in the first pair by the same factor.

8 [F-ADD] | student p. 115

Answer: $\frac{5}{6}$

Reasoning: Rename both fractions with a common denominator. Add the counts of equal parts.

9 [F-MULT] | student p. 115

Answer: $\frac{3}{5}$

Reasoning: An area model shows 12 of 20 equal parts, or three fifths.

10 [F-DIV] | student p. 115

Answer: It asks what fraction of a five-sixths group fits in one third. The answer is $\frac{2}{5}$.

Watch for $\frac{5}{18}$: He uses a familiar fraction rule without identifying the group.